

Name:

Category: CHEMISTRY

1. What is the mass percentage of nitrogen in ammonium nitrate ?

(N = 14 g/mol, H = 1 g/mol, O = 16 g/mol)

Options:

- A. 17.51%
- B. 28.01%
- C. 35.00%
- D. 42.44%

2. A sample of a pure iron oxide compound weighs 15.97 g and is found to contain 11.17 g of iron. What is the mass percentage of oxygen in this compound?

Options:

- A. 30.06%
- B. 69.94%
- C. 45.00%
- D. 25.12%

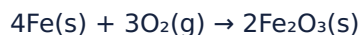
3. When 5.00 g of hydrated magnesium sulfate ($\text{MgSO}_4 \cdot x\text{H}_2\text{O}$) is completely dehydrated, it yields 2.44 g of anhydrous MgSO_4 . Find the coefficient x.

(MgSO_4 = 120 g/mol, H_2O = 18. g/mol)

Options:

- A. 1
- B. 4
- C. 6
- D. 7

4. What mass of iron is required to completely react with excess oxygen to produce 31.94 g of iron(III) oxide ?



(Fe = 56 g/mol, Fe_2O_3 = 160 g/mol)

Options:

- A. 11.17 g
- B. 22.34 g
- C. 44.68 g
- D. 89.36 g

5. Calcium carbonate decomposes upon heating to form calcium oxide and carbon dioxide gas. What mass of CaCO_3 is needed to produce 14.02 g of CaO?

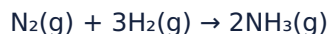


(CaCO_3 = 100 g/mol, CaO = 56 g/mol)

Options:

- A. 25 g
- B. 50. g
- C. 14. g
- D. 100 g

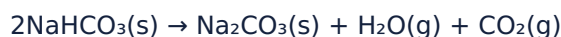
6. If 12.0 dm^3 of hydrogen gas reacts completely with excess nitrogen gas to form ammonia gas at the same temperature and pressure, what volume of NH_3 is produced?



Options:

- A. 4.0 dm^3
- B. 8.0 dm^3
- C. 12.0 dm^3
- D. 24.0 dm^3

7. What volume of carbon dioxide gas, measured at STP, is released when 2.00 moles of sodium bicarbonate decomposes completely?

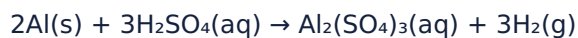


(Molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

Options:

- A. 11.2 dm^3
- B. 22.4 dm^3
- C. 44.8 dm^3
- D. 89.6 dm^3

8. What volume of hydrogen gas at STP is produced when 5.40 g of aluminum reacts completely with excess sulfuric acid?



(Al = 27 g/mol. Molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

Options:

- A. 2.24 dm^3
- B. 4.48 dm^3
- C. 6.72 dm^3
- D. 13.44 dm^3

9. A given mass of gas occupies a volume of 4.0 dm^3 at a pressure of 600 mmHg and a temperature of 27°C . If the gas is heated to 177°C while its pressure is simultaneously increased to 900 mmHg, calculate its new volume.

Options:

- A. 2.0 dm^3
- B. 3.0 dm^3
- C. 4.0 dm^3
- D. 6.0 dm^3

10. At a constant temperature, a sample of gas has a volume of 6.0 dm^3 when the pressure is 800 mmHg. If the volume is decreased by 25% of its original value, what will be the final pressure of the gas?

Options:

- A. 600 mmHg
- B. 1000 mmHg
- C. 1066.7 mmHg
- D. 1200 mmHg

11. A fixed mass of gas occupies 9.0 dm^3 at 127°C under a constant pressure. To what temperature in $^\circ\text{C}$ must the gas be cooled so that its new volume becomes 6.0 dm^3 ?

Options:

- A. -6.3°C
- B. 5.0°C
- C. 84.7°C
- D. 266.7°C

12. A gas trapped in a cylinder with a movable piston has a volume of 3.5 dm^3 at a pressure of 760 mmHg and a temperature of 0°C . What will the volume become if the pressure drops to 380 mmHg and the temperature rises to 273°C ?

Options:

- A. 1.75 dm^3
- B. 7.0 dm^3
- C. 14.0 dm^3
- D. 28.0 dm^3

13. A container holds a specific mass of gas at 47°C and 720 mmHg . If the gas undergoes a process where its volume is cut in half and its absolute temperature is doubled, calculate the final pressure.

Options:

- A. 360 mmHg
- B. 1440 mmHg
- C. 2160 mmHg
- D. 2880 mmHg

14. Under a constant pressure of 750 mmHg , a gas balloon has an initial volume of 2.5 dm^3 at a temperature of 25°C . If the temperature is increased to 144°C , what is the increase in the volume of the balloon?

Options:

- A. 1.0 dm^3
- B. 2.5 dm^3
- C. 3.5 dm^3
- D. 14.4 dm^3

15. At a constant temperature, the pressure of 8.0 dm^3 of gas is increased from 640 mmHg to 960 mmHg . Calculate the volume of gas expelled from the container.

Options:

- A. 2.67 dm^3
- B. 4.0 dm^3
- C. 5.33 dm^3
- D. 12.0 dm^3

16. A gas sample has a volume of 12.0 dm^3 at 27°C and a certain pressure. If the pressure is doubled and the temperature is dropped to -23°C , what is the final volume?

Options:

- A. 5.0 dm^3
- B. 6.0 dm^3
- C. 10.0 dm^3
- D. 20.0 dm^3

17. A weather balloon contains 15.0 dm^3 of gas at a pressure of 760 mmHg and a temperature of 30°C . At a high altitude, the volume expands to 40.0 dm^3 and the temperature drops to -31°C . What is the pressure at this altitude?

Options:

- A. 182.4 mmHg
- B. 228.0 mmHg
- C. 304.0 mmHg
- D. 418.5 mmHg

18. The gas commonly used for filling aircraft tyres is

Options:

- A. Oxygen
- B. Hydrogen
- C. Nitrogen
- D. Carbon dioxide

19. Rust is best represented by the formula

Options:

- A. $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- B. $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- C. Fe_3O_4
- D. $\text{Fe}(\text{OH})_2$

20. The process of coating iron with zinc to prevent corrosion is called

Options:

- A. Electrolysis
- B. Galvanization
- C. Vulcanization
- D. Anodization

21. The chemical formula of household bleach is

Options:

- A. NaCl
- B. NaClO
- C. $\text{Ca}(\text{OCl})_2$
- D. HClO_4

22. Which of the following is the strongest oxidizing agent?

Options:

- A. Cl_2
- B. Br_2
- C. I_2
- D. F_2

23. Graphite conducts electricity mainly because it contains

Options:

- A. Mobile ions
- B. Free mobile electrons
- C. Free protons
- D. Neutrons

24. The industrial manufacture of ammonia is carried out by the

Options:

- A. Bosch process
- B. Contact process
- C. Ostwald process
- D. Haber process

25. The catalyst used in the Haber process is

Options:

- A. Nickel
- B. Vanadium(V) oxide
- C. Finely divided iron
- D. Platinum

26. How many π bonds are present in a benzene molecule?

Options:

- A. 1
- B. 3
- C. 6
- D. 12

27. The most characteristic reaction of benzene is

Options:

- A. Nucleophilic addition
- B. Electrophilic substitution
- C. Free-radical substitution
- D. Electrophilic addition

28. The bond angle in ammonia is approximately

Options:

- A. 90°
- B. 104.5°
- C. 107°
- D. 109.5°

29. How many lone pairs of electrons are present in a water molecule?

Options:

- A. 1
- B. 2
- C. 3
- D. 4

30. Which gas law has the same dimensions as work done

Options:

- A. Boyle's law
- B. Charles' law
- C. Pascals law
- D. Dalton's law

31. Bicycle pump operate on what principle/law

Options:

- A. Charles' law
- B. Pressure law
- C. Boyle's law
- D. Archenemies principles

32. Which of the following gas mixtures at room temperature obeys Dalton's law of partial pressures?

Options:

- A. NO and O₂
- B. CO and CO₂
- C. NH₃ and HCl
- D. SO₂ and O₂

33. Three gases are held in a container: oxygen, carbon dioxide, and helium. The three gases have partial pressures of 2.00 atm, 3.00 atm, and 4.00 atm, respectively. How much is the total pressure inside the container?

Options:

- A. 6 atm
- B. 8 atm
- C. 9 atm
- D. 12 atm

34. At -18°C, a 2dm³ mixture of helium, nitrogen, and neon has a total pressure of 815 mmHg. What mass of neon is present in the mixture if the partial pressure of helium is 201 mmHg and the partial pressure of nitrogen is 351 mmHg? (Ne=20g/mol , He = 4g/mol, N₂=28g/mol).

Options:

- A. 0.33 g
- B. 0.66 g
- C. 22.4 g
- D. 4.00 g

35. A container containing two gases, helium and argon, contains 30.0% helium by volume. If the total pressure inside the container is 4.00 atm, calculate the partial pressures of argon.

Options:

- A. 1.2 atm
- B. 2.2 atm
- C. 2.8 atm
- D. 5.6 atm

36. A flask contains 14.0 g of hydrogen, 84.0 g of nitrogen, and 2.00 moles of oxygen. What is the total pressure in the flask if the partial pressure of oxygen is 78.00 mm of mercury?

Options:

- A. 468 mm Hg
- B. 764 mm Hg
- C. 936 mmHg
- D. 760 mm Hg

37. The partial pressure of oxygen in a sample of air is 452 mm Hg and the total pressure is 780 mm Hg. What is the mole fraction of oxygen?

Options:

- A. 0.203
- B. 0.579
- C. 2.030
- D. 5.790

38. In a closed system, a sample of gas A evaporates over water. What is the pressure of gas A if the total pressure is 780 torr and the pressure of water vapour is 1 atm?

Options:

- A. 1 atm
- B. 0.03 atm
- C. 0.3 atm
- D. 3.0 atm

39. At 50.0 °C, 80 dm³ of oxygen are collected over water. The room has an atmospheric pressure of 96.00 kPa. Determine the partial pressure of oxygen

Options:

- A. 0.83 atm
- B. 2.82 atm
- C. 2.24 atm
- D. 1.20 atm

40. At a total pressure of 10 atm, 56 g of nitrogen and 96 g of oxygen are mixed isothermally. The ratio of oxygen and nitrogen partial pressures is

Options:

- A. 3:2
- B. 2:3
- C. 3:1
- D. 2:1

Category: PHYSICS

1. A radioactive sample has an initial activity of 800 counts per minute. If the decay constant of the element is 0.05775 day⁻¹, calculate the remaining activity of the sample after 24 days.

Options:

- A. 50 counts per minute
- B. 100 counts per minute
- C. 200 counts per minute
- D. 400 counts per minute

2. An unstable isotope X with a mass number of 238 and atomic number of 92 undergoes a series of radioactive decays by emitting 3 alpha (α) particles and 2 beta (β^-) particles. Determine the mass number (A) and atomic number (Z) of the resulting stable daughter nucleus Y.

Options:

- A. A = 226, Z = 88
- B. A = 226, Z = 86
- C. A = 232, Z = 88
- D. A = 232, Z = 86

3. A radioactive isotope of mass 64 g has a half-life of 4 days. What mass of the substance will disintegrate after a total period of 16 days?

Options:

- A. 4 g
- B. 16 g
- C. 48 g
- D. 60 g

4. A radioactive tracer sample has a half-life of 3 hours. If it is recorded that only $\frac{1}{32}$ of the original mass of the sample remains undecayed, determine the total time that has elapsed.

Options:

- A. 9 hours
- B. 12 hours
- C. 15 hours
- D. 18 hours

5. The count rate of a sample of radioactive material falls from 1600 counts per second to 200 counts per second in 12 minutes. Calculate the decay constant (λ) of this element.

Options:

- A. 0.173 min^{-1}
- B. 0.231 min^{-1}
- C. 0.347 min^{-1}
- D. 0.693 min^{-1}

6. A radioactive element has an initial mass of 120 g. If $\frac{7}{8}$ of the sample disintegrates in 24 hours, find its half-life.

Options:

- A. 3 hours
- B. 6 hours
- C. 8 hours
- D. 12 hours

7. The half-life of a specific nuclide is 8 days. What fraction of the sample will remain undecayed after 32 days?

Options:

- A. $\frac{1}{4}$
- B. $\frac{1}{8}$
- C. $\frac{1}{16}$
- D. $\frac{1}{32}$

8. A pure sample of a radioactive substance has a half-life of 10 seconds. If the detector records a count rate of 20 counts per minute from the background radiation alone, what will a detector read next to a sample with an initial activity of 320 counts per minute after 30 seconds have passed?

Options:

- A. 40 counts per minute
- B. 60 counts per minute
- C. 80 counts per minute
- D. 100 counts per minute

9. A radioactive mass decreases from 80 g to 5 g within a timeframe of 20 days. Determine the decay constant (λ) of this isotope.

Options:

- A. 0.0693 day^{-1}
- B. 0.1386 day^{-1}
- C. 0.2772 day^{-1}
- D. 0.5544 day^{-1}

10. A radioactive nucleus undergoes natural transmutation according to the nuclear equation below:



Evaluate the quantities x and y.

Options:

- A. $x = 2, y = 2$
- B. $x = 2, y = 1$
- C. $x = 4, y = 2$
- D. $x = 2, y = 0$

11. Uranium-238 with atomic number 92 emitted 8 alpha particles, 4 beta particles, what will be the ratio of parent's element and the new element's proton number

Options:

- A. 4:5
- B. 23:20
- C. 1:2
- D. 2:1

12. The radioactive disintegration count rate of a sample drops from 800 counts per minute to 100 counts per minute in exactly 15 hours. What is the half-life of this radioactive element?

Options:

- A. 3 hours
- B. 5 hours
- C. 7.5 hours
- D. 30 hours

13. A certain radioactive isotope has a half-life of 4 hours. What is the value of its decay constant (λ) in reciprocal seconds (s^{-1})?

Options:

- A. $4.81 \times 10^{-5} \text{ s}^{-1}$
- B. $2.77 \times 10^{-4} \text{ s}^{-1}$
- C. $6.93 \times 10^{-2} \text{ s}^{-1}$
- D. $1.73 \times 10^{-1} \text{ s}^{-1}$

14. What type of nuclear reaction is represented by the equation:



Options:

- A. Nuclear fission
- B. Nuclear fusion
- C. Artificial transmutation
- D. Alpha decay

15. The nuclear radiation which is deflected to the negative side of an electric field is

Options:

- A. Beta particles
- B. Alpha particles
- C. X-ray
- D. Gamma rays

16. The nuclear radiation which is electromagnetic in nature is

Options:

- A. Alpha particles
- B. Beta particles
- C. Gamma rays
- D. X-rays

17. A radioactive sample has a half-life of 8 hours. What percentage of the sample that remain undecayed after 24 hours?

Options:

- A. 87.5%
- B. 12.5%
- C. 25%
- D. 75%

18. Arrange them in order of increasing penetration power

Options:

- A. Alpha > Beta > Gamma
- B. Gamma < Beta < Alpha
- C. Alpha < beta < Gamma
- D. Alpha < Gamma < Beta

19. Which of these is not a nuclear radiation

Options:

- A. Gamma rays
- B. X-rays
- C. Beta particles
- D. Alpha particles

20. A radioactive element decays to one-sixteenth of its original mass in 24 minutes. Calculate the mean life of the element.

Options:

- A. 1.5 minutes
- B. 6.0 minutes
- C. $6/\ln 2$ minutes
- D. $1.5/\ln 2$ minutes

21. Which of the following is used to make waves appear stationary

Options:

- A. Ripple
- B. Stroboscope
- C. Plunger
- D. Polaroids

22. A point in a longitudinal waves where particles are spread far apart is ??

Options:

- A. Node
- B. Compression
- C. Rarefactions
- D. Antinode

23. The distance between 2 nodes in a stationary wave is?

Options:

- A. One wavelength
- B. Half wavelength
- C. $\frac{2}{3}$ wavelength
- D. 2 wavelength

24. Calculate the speed of sound in copper whose density and Young's modulus are $8.9 \times 10^3 \text{kg/m}^3$ and $1.2 \times 10^{11} \text{N/m}^2$ respectively.

Options:

- A. 3670m/s
- B. 3672m/s
- C. 4000m/s
- D. 3000m/s

25. In a destructive interference, the 2 waves are how many degrees out of phase??

Options:

- A. 90°
- B. 100°
- C. 180°
- D. 360°

26. When an unpolarized light is polarized, the intensity??

Options:

- A. Reduces to quarter of its original
- B. Doubles
- C. Halves
- D. Remains constant

27. Which of the following is mechanical and transverse??

Options:

- A. Sound waves
- B. Light waves
- C. Water waves
- D. Electromagnetic waves

28. Audible frequency is between??

Options:

- A. 30HZ to 30KHZ
- B. 20HZ to 30KHZ
- C. 20HZ to 20KHZ
- D. 20HZ to 23.5KHZ

29. The ceilings of Auditoriums and halls are covered with boards to reduce??

Options:

- A. Reflection of sound
- B. Reverberation of sound
- C. Quality of sound
- D. Loudness of sound

30. The pitch of a note depends on ??

Options:

- A. Harmonics
- B. Overtone
- C. Frequency
- D. Intensity

31. What is the S.I unit of intensity

A.Loudness

Options:

- A. Joules
- B. Decibels
- C. Cannot be measured

32. The quality of a note depends on??

Options:

- A. Frequency
- B. Wavelength
- C. Harmonics
- D. Intensity

33. A boy performed an experiment to determine the speed of sound in air Using the resonance column method. The length of the air column that resonates in the fundamental mode with a tuning fork is 0.2m. if the length is varied such that the same fork resonates with the first overtone at 0.7m. calculate the end correction

Options:

- A. 0.5m
- B. 0.05m
- C. 50cm
- D. 0.9m

34. ____ is an interval whose higher note has a frequency twice that of its lower note

Options:

- A. Overtone
- B. Beat
- C. Octave
- D. Resonance

35. Which of the following is a percussion instrument??

Options:

- A. Flute
- B. Guitar

- C. Drums
- D. Violin

36. 60 complete waves pass a particular point in 4 seconds. If the distance between 3 successive troughs of the wave is 15 m , calculate the speed of the wave.

Options:

- A. 113.5m/s
- B. 112.5m/s
- C. 16m/s
- D. 112.5m/s²

37. The distance between two successive crests of a water wave is 1m. If a particle on the surface of the water makes two complete vertical oscillations in 1s. calculate the speed of the wave .

Options:

- A. 3m/s
- B. 2m/s
- C. 15m/s
- D. 8m/s

38. A man clapping his hands at regular intervals observes that the echoes of a clap coincides with the next clap. If the reflecting cliff is 160m away and the speed of sound is 320m/s, what is the frequency of the clapping??

Options:

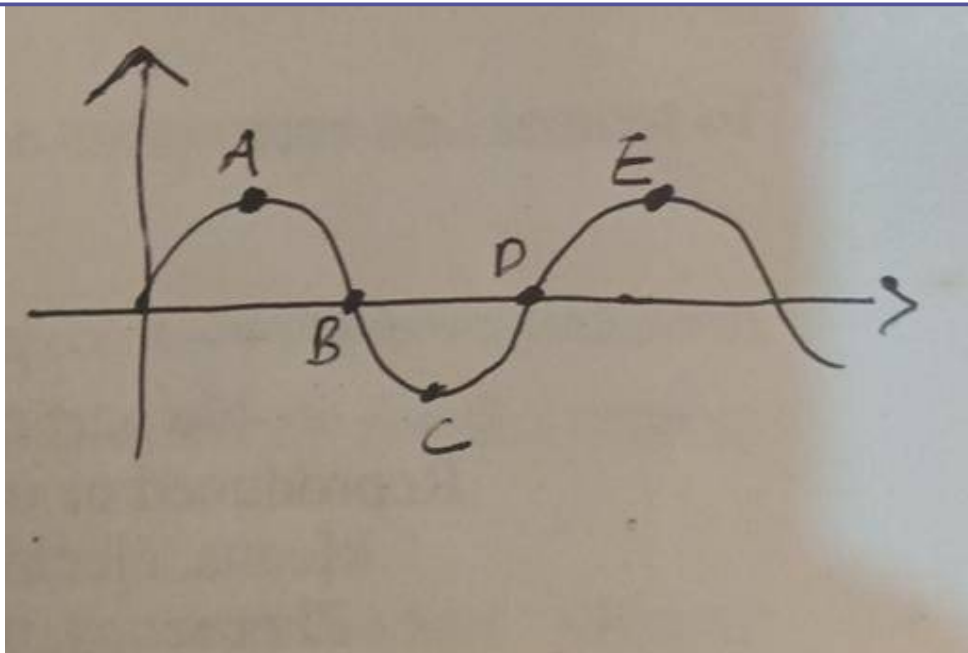
- A. 1s⁻¹
- B. 10HZ
- C. 5HZ
- D. No answer

39. Calculate the frequency of the first overtone of a closed pipe of length 40cm if the velocity of the sound waves generated in the pipe when it gives its fundamental more is 360m/s.

Options:

- A. 675Hz
- B. 225Hz
- C. 675s
- D. 241Hz

40.



Which of the following points are in phase

Options:

- A. A and B
- B. B and D
- C. A and E
- D. C and E

Category: USE OF ENGLISH

1. (Choose the option that correctly completes the sentence)

The _____ common room has been renovated with modern furniture.

Options:

- A. lecturer's
- B. lecturers's
- C. lecturers'
- D. lecturers

2. (Choose the option that correctly completes the sentence)

_____ and _____ grades showed a significant improvement this term.

Options:

- A. Chidi's / Tunde's
- B. Chidi / Tunde's
- C. Chidi's / Tunde
- D. Chidis / Tundes

3. (Choose the option that correctly completes the sentence)

You need a _____ rest before sitting for the intensive three-hour exam.

Options:

- A. good nights'
- B. good night's
- C. good nights
- D. good night

4. (Choose the option that correctly completes the sentence)

The security guard found a _____ wallet on the floor of the reception hall.

Options:

- A. gentlemen's
- B. gentlemen
- C. gentleman's
- D. gentleman

5. (Choose the option that correctly completes the sentence)

The office belongs to the two managing directors; it is the _____ office.

Options:

- A. director's
- B. directors'
- C. directors's
- D. directors

6. (Choose the option that correctly completes the sentence)

_____ a shame that the team lost the final match after training so hard.

Options:

- A. Its
- B. It's
- C. Its'
- D. Itis'

7. (Choose the option that correctly completes the sentence)

My two _____ houses are located in the same residential estate in Lekki.

Options:

- A. brother-in-law's
- B. brothers-in-law's
- C. brothers'-in-law
- D. brother's-in-law

8. (Choose the option that correctly completes the sentence)

Are you sure this modern laptop is _____?

Options:

- A. your's
- B. yours'
- C. yours
- D. you're

9. (Choose the option that correctly completes the sentence)

The _____ committee submitted its comprehensive report to the Vice-Chancellor.

Options:

- A. alumna's
- B. alumnis'
- C. alumni's
- D. alumni

10. (Choose the option that correctly completes the sentence)

The young boy always remembers to dot his _____ and cross his _____.

Options:

- A. i's / t's
- B. is / ts
- C. i/s / t/s
- D. is' / ts'

11. (Choose the option that correctly completes the sentence)

The _____ association held its annual conference yesterday.

Options:

- A. teacher's
- B. teachers'
- C. teachers
- D. teachers's

12. (Choose the option that correctly completes the sentence)

_____ laptop was left in the chemistry laboratory.

Options:

- A. Charles's
- B. Charles'
- C. Charles
- D. Both A and B are correct

13. (Choose the option that correctly completes the sentence)

The children found a _____ nest in the mango tree.

Options:

- A. birds'
- B. bird's
- C. birds
- D. birds's

14. (Choose the option that correctly completes the sentence)

_____ going to be a very long journey to Abuja.

Options:

- A. Its'
- B. It's
- C. Its
- D. Itis'

15. (Choose the option that correctly completes the sentence)

The principal inspected the _____ dormitory during his routine check.

Options:

- A. boy's
- B. boys's
- C. boys'
- D. boys

16. (Choose the option that correctly completes the sentence)

Amadi and Obi bought a car together; that vehicle is _____.

Options:

- A. Amadi's and Obi's

- B. Amadi and Obi's
- C. Amadi and Obis'
- D. Amadis and Obi's

17. (Choose the option that correctly completes the sentence)

The _____ bathroom is located on the ground floor of the mall.

Options:

- A. women's
- B. womens'
- C. womens
- D. womens's

18. (Choose the option that correctly completes the sentence)

The cat licked _____ injured paw after escaping from the trap.

Options:

- A. it's
- B. its'
- C. its
- D. itself's

19. (Choose the option that correctly completes the sentence)

We need to buy two _____ worth of fuel for the generator.

Options:

- A. thousand nairas'
- B. thousand nairas
- C. thousand naira
- D. thousand naira's

20. (Choose the option that correctly completes the sentence)

This book belongs to my sister; it is _____.

Options:

- A. her's
- B. hers
- C. hers'
- D. her

21. Between you and _____, I think the plan will fail.

Options:

- A. I
- B. me
- C. myself
- D. mine

22. One must respect _____ parents and elders.

Options:

- A. his
- B. their
- C. one's
- D. her

23. The principal punished the boys, each of _____ was late.

Options:

- A. whose
- B. who
- C. whom
- D. which

24. Ade and Bimpe are fond of _____.

Options:

- A. each other
- B. one another
- C. themselves
- D. each-themselves

25. _____ of the two candidates qualified for the scholarship.

Options:

- A. Neither
- B. All
- C. None
- D. Both

26. She is taller than _____.

Options:

- A. I
- B. me
- C. mine
- D. myself

27. The dog hurt _____ when it jumped over the fence.

Options:

- A. it
- B. itself
- C. himself
- D. theirself

28. _____ do you think I met at the bookstore?

Options:

- A. Who
- B. Whom
- C. Whose
- D. Which

29. Neither the teacher nor the students brought _____ books.

Options:

- A. his
- B. her
- C. their
- D. our

30. It was _____ who won the first prize in the competition.

Options:

- A. whom
- B. him
- C. me
- D. he

31. The girl _____ mother won the lottery has traveled abroad.

Options:

- A. who
- B. whom
- C. whose
- D. which

32. The jury could not agree on _____ verdict.

Options:

- A. its
- B. their
- C. it's
- D. theirs

33. It's time Chidi and _____ left for the examination hall.

Options:

- A. me
- B. I
- C. myself
- D. mine

34. To _____ was the letter addressed?

Options:

- A. who
- B. whom
- C. whose
- D. which

35. The cookies were shared among the four siblings, and each took _____ share.

Options:

- A. his
- B. their
- C. its
- D. theirs

36. We should help _____ in times of trouble, as members of the same church.

Options:

- A. each other
- B. one another
- C. ourselves
- D. each-other

37. Of the three sisters, Obiageli is the one _____ I think will pass.

Options:

- A. who
- B. whom
- C. whose
- D. which

38. The car over there is _____.

Options:

- A. our's
- B. ours
- C. our
- D. ours'

39. Either the boys or their father has forgotten _____ keys.

Options:

- A. their
- B. his
- C. our
- D. its

40. Let you and _____ settle this matter amicably.

Options:

- A. I
- B. me
- C. myself
- D. mine

41. The boy _____ you accused of stealing has an alibi.

Options:

- A. who
- B. whom
- C. whose
- D. which

42. Every student must bring _____ own calculator to the physics exam.

Options:

- A. they're
- B. there
- C. their
- D. his

43. The responsibility for organizing the event lies with you and _____.

Options:

- A. he
- B. him
- C. himself
- D. his

44. Five candidates applied for the post, but _____ of them met the requirements.

Options:

- A. neither
- B. none
- C. no one
- D. nobody

45. If I were _____, I would register for the evening classes.

Options:

- A. him
- B. he
- C. himself
- D. his

46. The dynamic duo promised to stand by _____ during the trial.

Options:

- A. each other
- B. one another
- C. themselves
- D. each-other

47. Uncle Tunde bought a smartphone for my brother and _____ as graduation gifts.

Options:

- A. I
- B. me
- C. myself
- D. mine

48. That beautiful house on the corner is _____.

Options:

- A. their's
- B. theirs'
- C. theirs
- D. there's

49. The cat licked _____ paws after eating the fish.

Options:

- A. its'
- B. it's
- C. its
- D. itself

50. _____ did the vice-chancellor say should head the committee?

Options:

- A. Whom
- B. Who
- C. Whose
- D. Which

51. (Choose the option nearest in meaning to the word in quotation marks)

The manager's "lucid" explanation helped the team complete the project ahead of schedule.

Options:

- A. ambiguous
- B. clear
- C. lengthy
- D. complicated

52. (Choose the option nearest in meaning to the word in quotation marks)

Despite the heavy security, the thief managed to "elude" the police.

Options:

- A. confront
- B. assist
- C. evade
- D. attack

53. (Choose the option nearest in meaning to the word in quotation marks)

The politician was criticized for making "frivolous" remarks during the serious national debate.

Options:

- A. profound
- B. trivial
- C. deliberate
- D. crucial

54. (Choose the option nearest in meaning to the word in quotation marks)

The sudden rainstorm "marred" our plans for an outdoor graduation party.

Options:

- A. ruined
- B. enhanced
- C. delayed
- D. sustained

55. (Choose the option nearest in meaning to the word in quotation marks)

Young children are often highly "gullible" and will believe almost any story you tell them.

Options:

- A. shrewd
- B. stubborn
- C. naive
- D. cynical

56. (Choose the option opposite in meaning to the word in quotation marks)

The witness gave a "voluntary" statement to the police concerning the accident.

Options:

- A. deliberate
- B. compulsory
- C. intentional
- D. optional

57. (Choose the option opposite in meaning to the word in quotation marks)

Unlike his brother who is highly "loquacious", Tunde rarely speaks during family gatherings.

Options:

- A. talkative

- B. eloquent
- C. reserved
- D. arrogant

58. (Choose the option opposite in meaning to the word in quotation marks)

The economic policies implemented by the new administration have caused a "scarcity" of essential commodities.

Options:

- A. shortage
- B. deficit
- C. abundance
- D. poverty

59. (Choose the option opposite in meaning to the word in quotation marks)

After hours of intense negotiations, the two warring factions finally reached an "amicable" settlement.

Options:

- A. hostile
- B. peaceful
- C. friendly
- D. abrupt

60. (Choose the option opposite in meaning to the word in quotation marks)

The teacher praised the student for being "diligent" in her academic duties.

Options:

- A. industrious
- B. intelligent
- C. lazy
- D. careful

Category: BIOLOGY

1. Which part of the brain controls body balance?

Options:

- A. Cerebrum
- B. Cerebellum
- C. Thalamus
- D. Medulla

2. What is the outermost layer of the meninges?

Options:

- A. Pia mater
- B. Dura mater
- C. Arachnoid mater
- D. Neurilemma

3. Sensory neurons enter the spinal cord through the

Options:

- A. Ventral root
- B. Dorsal root
- C. Central canal

D. Grey horn

4. Which brain part controls heartbeat and breathing?

Options:

- A. Cerebrum
- B. Cerebellum
- C. Medulla oblongata
- D. Hypothalamus

5. The left and right cerebral hemispheres are joined by the

Options:

- A. Pons Varolii
- B. Corpus callosum
- C. Spinal cord
- D. Meninges

6. Which structure acts as the body's thermostat?

Options:

- A. Hypothalamus
- B. Thalamus
- C. Cerebrum
- D. Medulla

7. Shock absorption in the CNS is provided by

Options:

- A. Blood
- B. Lymph
- C. Cerebrospinal fluid
- D. Endolymph

8. Motor neurons leave the spinal cord through the

Options:

- A. Dorsal root
- B. Ventral root
- C. Lateral root
- D. Ganglion

9. Which part of the brain controls voluntary thinking?

Options:

- A. Cerebellum
- B. Cerebrum
- C. Medulla
- D. Hypothalamus

10. The main light spectrum captured by chlorophyll during photosynthesis is/are.....?

Options:

- A. Green and yellow
- B. Blue and red
- C. Orange and violet
- D. Green only

11. The most important pigment in the chloroplast for photosynthesis is.....?

Options:

- A. Carotenoid
- B. Xanthophyll
- C. Chlorophyll a
- D. Chlorophyll b

12. Photolysis of water occurs at what stage of photosynthesis?

Options:

- A. Dark reaction
- B. Light reaction
- C. Calvin cycle
- D. Glycolysis

13. Dodder and are typical examples of parasitic plants.

Options:

- A. Mistletoe
- B. Pitcher plant
- C. Venus flytrap
- D. Sundew

14. The energy battery of a living cell is called.....?

Options:

- A. Nucleus
- B. ADT
- C. ATP
- D. Mitochondria

15. Vitamin B12 is also known chemically as

Options:

- A. Thiamine
- B. Cyanocobalamin
- C. Riboflavin
- D. Niacin

16. The most important adaptive feeding structure of a parasitic plant is the

Options:

- A. Rhizoid
- B. Haustorium
- C. Taproot
- D. Tendril

17. The optimum temperature for most human enzymes to function is approximately

Options:

- A. 0°C to 10°C
- B. 37°C
- C. 60°C
- D. 98°C

18. Lack of roughage/fiber in a human diet leads directly to

Options:

- A. Diarrhoea
- B. Constipation
- C. Kwashiorkor
- D. Scurvy

19. Pellagra is a deficiency symptom of which vitamin?

Options:

- A. Vitamin B1
- B. Vitamin B3
- C. Vitamin C
- D. Vitamin D

20. Which mineral deficiency leads to muscle paralysis?

Options:

- A. Iron
- B. Iodine
- C. Potassium
- D. Calcium

21. Deficiency of cobalt causes

Options:

- A. Pernicious anemia
- B. Scurvy
- C. Night blindness
- D. Beriberi

22. The main chemical components of a Biuret test are

Options:

- A. Copper (II) sulfate and Sodium hydroxide
- B. Iodine solution and Potassium iodide
- C. Ethanol and Water
- D. Sudan III and Hydrochloric acid

23. The process of breaking down glucose in the absence of oxygen to produce lactic acid or ethanol is called

Options:

- A. Aerobic respiration
- B. Fermentation
- C. Photosynthesis
- D. Transpiration

24. Which plant hormone is primarily responsible for fruit ripening?

Options:

- A. Cytokinin
- B. Ethylene
- C. Auxin
- D. Gibberellin

25. The loss of water vapor from the aerial parts of a plant, mainly through the stomata, is known as

Options:

- A. Osmosis
- B. Translocation
- C. Transpiration
- D. Guttation

26. Which vitamin is essential for blood clotting, and its deficiency leads to prolonged bleeding?

Options:

- A. Vitamin A
- B. Vitamin C
- C. Vitamin K
- D. Vitamin D

27. Which mineral is an essential component of hemoglobin, and its deficiency causes nutritional anemia?

Options:

- A. Manganese
- B. Iron
- C. Magnesium
- D. Phosphorus

28. Dark stage reaction of photosynthesis occurs where?

Options:

- A. Grana
- B. Stroma
- C. Cell wall
- D. Ribosome

29. Light stage reaction of photosynthesis is also known as.....?

Options:

- A. Calvin cycle
- B. Hill's reaction
- C. Krebs cycle
- D. Glycolysis

30. Dark stage reaction of photosynthesis is also known as?

Options:

- A. Light-dependent reaction
- B. Photophosphorylation
- C. Calvin cycle
- D. Photolysis

31. The enzyme found in saliva is called....?

Options:

- A. Pepsin
- B. Lipase
- C. Ptyalin
- D. Trypsin

32. The vascular tissue of plants which also serves as a supporting unit is?

Options:

- A. Xylem
- B. Phloem
- C. Cambium
- D. Cortex

33. The main mineral element which aids in the formation of chlorophyll is?

Options:

- A. Iron
- B. Magnesium
- C. Calcium
- D. Nitrogen

34. Water culture is also known as.....?

Options:

- A. Hydroponics
- B. Aerophobic
- C. Amnesiac
- D. Algal blooms

35. Nitrogen nutrients needed by plants are required in the form of....

Options:

- A. Nitrates
- B. Nitrogen gas
- C. Amino acids directly
- D. Nitric oxide

36. Which of the following foods has the highest calorific value per gram?

Options:

- A. Rice
- B. Cheese
- C. Milk
- D. Beans

37. Which of the following organisms exhibits holozoic nutrition?

Options:

- A. Mushroom
- B. Amoeba
- C. Rhizopus
- D. Mucor

38. In saprophytic nutrition, digestion takes place

Options:

- A. inside the nucleus
- B. inside a food vacuole
- C. outside the organism's body
- D. inside the bloodstream

39. Which of the following is an example of a mutualistic association?

Options:

- A. Tapeworm and man
- B. Tick and dog
- C. Fungus and alga in lichen
- D. Mosquito and man

40. Which adaptation enables a tapeworm to remain attached to the intestine of its host?

Options:

- A. Presence of chloroplasts
- B. Possession of hooks and suckers
- C. Presence of a well-developed alimentary canal
- D. Presence of cilia